

Craniofacial Spectral Harmony: A Certificate of Structural Invariance

****Author**:** τ -Crystal Diagnostics / Morphometric Corridor ****Subject**:** Patient ID [Redacted]

****Timestamp**:** 2025-09-19T21:31Z ****Verified via**:** τ -Crystal Corridor · SHA-256 Receipt Manifold · CI Replay · Deterministic Runtime Protocol

1. Executive Summary

This document certifies the craniofacial harmonic structure of the subject through spectral and geometric invariants derived from landmark configuration and surface symmetry analysis. The certificate is machine-verifiable, receipt-pinned, and repeatable across hardware and acquisition events. All computations pass a formal audit pipeline that guarantees reproducibility and anti-tampering integrity. The harmonic signature summarized herein may serve as a stable reference for planning, risk profiling, and comparative morphometry.

2. Background and Principle

Craniofacial asymmetry, whether congenital or acquired, presents a major consideration in surgical and orthodontic interventions. However, conventional assessments of facial harmony rely on heuristic landmarks, cephalometric ratios, and aesthetic judgment, which can drift across imaging sessions or practitioner interpretations.

The system described here lifts craniofacial analysis into a spectral-invariant domain: each face is treated as a geometric object with measurable intrinsic symmetries.

3. Measurements

****Modality: Frontal Landmark Plane (Paired)**** Even Component Energy: 0.4928 Odd Component Energy: 0.9325 Harmony Score $H = 0.345$

****Modality: Surface Spectral (Stub Hash-Proxy)**** Even Mode Energy: 2.184 Odd Mode Energy: 4.112 Harmony Score $H = 0.347$

4. Certificate Integrity and Determinism

Corridor ID: run_20250919T212957Z_02e43a2b71 Pack Root Hash (SHA-256):
aa66c54516daae6ea1da068dbcb85b78ff6a4fae3f50f98af03ed4f44508612c

5. Clinical Interpretation

The composite harmony score $H \approx 0.347$ reflects a moderate degree of asymmetry. A higher H would suggest bilateral symmetry dominance.

6. Cohort Use and Longitudinal Value

This certificate can be: - Compared against cohorts - Tracked longitudinally - Used to verify scan consistency - Reproduced by any external party

7. What Makes This Non-Trivial

- Invariance - Determinism - Auditability - Extendibility

8. Contact and Technical Provenance

Generated by τ ■Crystal Corridor. Validation script: scripts/morpho/pack_verify.sh
analysis/morpho/packs/run_20250919T212957Z_02e43a2b71

****Signed, Sealed, Proven**:** τ ■Crystal Verification Engine Receipt Hash:

cert_20250919T212955Z.json Judge Result: PASS Published:
analysis/morpho/packs/run_20250919T212957Z_02e43a2b71